Gravitational Coupling to Entanglement Entropy Density

A Falsifiable Extension of Thermodynamic Gravity to Quantum-Coherent Systems

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Abstract

We derive a dimensionally consistent coupling between entanglement entropy density and spacetime curvature, extending Jacobson's thermodynamic formulation of general relativity [1] to macroscopic quantum-coherent systems. The modified Einstein equation takes the form

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu} + \frac{\tilde{\kappa}}{k_{\mathrm{B}} \ln 2}\, S_{\mathrm{ent}}\, g_{\mu\nu},$$

where S_{ent} is the entanglement entropy density (bit/m^3) and $\tilde{\kappa}$ is a dimensionless coupling constant. Under the *Planck-scale holographic regulator hypothesis* — the assumption that laboratory-scale entanglement entropy couples to geometry through the same Planck-length regulation that governs horizon entropy — we obtain the estimate $\tilde{\kappa} = -1/4$. This is an *estimate under an explicit hypothesis*, not a derivation from first principles. Realistic systems exhibit environmental suppression $\tilde{\kappa}_{\mathrm{eff}} = -(1/4)\alpha_{\mathrm{screen}}$, where α_{screen} is an open parameter requiring derivation from open-quantum-system dynamics.

Under spherical symmetry, the modified Poisson equation gives an entanglement-induced acceleration correction

$$\Delta a(R) = -\frac{\tilde{\kappa}\, c^4}{8\pi k_{\mathrm{B}} \ln 2} \cdot \frac{M_{\mathrm{ent}}(< R)}{R^2},$$

where $M_{\mathrm{ent}}(< R)$ is the enclosed entanglement entropy. For bounded sources this scales as $1/R^2$; a $1/R$ law requires a spatially extended source profile with $\rho_{\mathrm{ent}} \propto 1/r^2$.

The signal magnitude is currently indeterminate because α_{screen} and the Planck-scale renormalization of $\tilde{\kappa}$ are unresolved. The falsification criterion, however, is independent of signal magnitude: if macroscopic quantum-coherent systems ($\geq 10^6$ entangled qubits) exhibit no anomalous stress-energy at pressure sensitivity $\Delta p < 10^{-6}$ Pa after ≥ 1000 runs across three or more independent platforms, then $|\tilde{\kappa}| < 10^{-15}$, ruling out laboratory-scale relevance.

Ontological constraints. Classical spacetime, standard quantum mechanics, no new particles or modified geometry — only a modified stress-energy source via entanglement entropy.

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1 Introduction: Thermodynamic Gravity and Its Extension

1.1 Jacobson's Result

In 1995, Jacobson showed that Einstein's field equations follow from the Clausius relation $\delta Q = T\,dS$ applied to local Rindler horizons [1]. For an observer accelerating with proper acceleration a, the local horizon temperature is the Unruh temperature

$$T_U = \frac{\hbar a}{2\pi c k_{\mathrm{B}}}, \tag{1}$$

and the entropy associated with a horizon area element dA is the Bekenstein–Hawking entropy

$$dS_{BH} = \frac{k_{\mathrm{B}} c^3}{4G\hbar}\,dA. \tag{2}$$

Applying $\delta Q = T_U\,dS_{BH}$ and identifying the energy flux with the stress-energy tensor recovers $G_{\mu\nu} = 8\pi G c^{-4} T_{\mu\nu}$ exactly.

Gravity is therefore not fundamental in this view: it is the thermodynamic equation of state of spacetime when entropy scales with horizon area.

1.2 The Natural Extension

If gravity emerges from area-law entanglement entropy at horizons, the question arises: does volume-law entanglement entropy in a laboratory quantum system contribute to the gravitational source term? The horizon is absent, but the entanglement is physically real.

We propose this as a falsifiable physical hypothesis. Supporting evidence includes Verlinde's 2017 demonstration that entropic gravity extends beyond the strict horizon case [2], and Bose et al.'s observation of gravity-mediated entanglement between massive quantum objects without causal horizons [3].

2 The Modified Einstein Equation: Dimensional Rigor

2.1 Bit-to-Entropy Conversion

"Bit" is a counting unit, dimensionless like "dozen." Physical entropy requires conversion:

$$S_{\mathrm{physical}} = I_{\mathrm{bits}} \cdot k_{\mathrm{B}} \ln 2, \qquad k_{\mathrm{B}} \ln 2 = 9.570 \times 10^{-24}\ \mathrm{J/K}. \tag{3}$$

We define:

$$S_{\mathrm{ent}} \equiv \text{entanglement entropy density } (\mathrm{bit/m^3}), \tag{4}$$

$$\mathcal{S}_{\mathrm{ent}} \equiv S_{\mathrm{ent}} \cdot k_{\mathrm{B}} \ln 2 \quad [\mathrm{J/(K{\cdot}m^3)}]\text{: physical entropy density.} \tag{5}$$

2.2 The Modified Field Equation

$$\boxed{G_{\mu\nu} = \frac{8\pi G}{c^4}\left(T_{\mu\nu} + \tilde{\kappa}\,\frac{c^4}{8\pi G k_{\mathrm{B}} \ln 2}\,S_{\mathrm{ent}}\,g_{\mu\nu}\right)} \tag{6}$$

Dimensional verification. The entanglement source term must match $[T_{\mu\nu}] = \mathrm{kg\,m^{-1}s^{-2}}$:

$$\left[\tilde{\kappa}\frac{c^4}{8\pi G k_{\mathrm{B}} \ln 2} S_{\mathrm{ent}}\right] = \frac{\mathrm{m^4 s^{-4}}}{\mathrm{m^3 kg^{-1} s^{-2}} \cdot \mathrm{kg\,m^2 s^{-2} K^{-1}}} \cdot \mathrm{K\,m^{-3}} = \mathrm{kg\,m^{-1} s^{-2}}.\checkmark \tag{7}$$

Repulsive gravity condition. For a perfect fluid with energy density ρ_m and pressure p:

$$\rho_{\mathrm{grav}} + \frac{3p_{\mathrm{grav}}}{c^2} = \rho_m + \frac{3p}{c^2} + \frac{3\tilde{\kappa}c^2}{8\pi G k_{\mathrm{B}} \ln 2}\,S_{\mathrm{ent}}. \tag{8}$$

For $\tilde{\kappa} < 0$ and $S_{\mathrm{ent}} > 0$, the entanglement term contributes negative effective pressure, producing repulsive curvature without exotic matter.

3 The Extrapolation Boundary

This section is the intellectual center of the paper.

3.1 Where Jacobson's Derivation Applies

Jacobson's result holds exactly under three conditions:

1. A causal horizon exists, defining a well-posed Unruh temperature (1).
2. Entanglement entropy scales with horizon *area* (Bekenstein–Hawking law).
3. The surface gravity κ_s is well defined and uniform on the horizon.

None of these conditions holds in a laboratory quantum system: there is no causal horizon, no Unruh temperature, and GHZ-state entanglement entropy scales with atom *number* (volume law), not surface area.

3.2 The Hypothesis

Hypothesis 3.1 (Holographic Regulator Extension)**.** The coupling between entanglement entropy density and spacetime curvature, which follows rigorously from area-law entropy at causal horizons, extends to systems with volume-law entanglement entropy in the absence of causal horizons, regulated by the same Planck-scale geometric structure that governs horizon area entropy.

Hypothesis 3.1 is motivated by holographic universality and the Bose et al. result [3], but it is not mathematically derivable from established physics. Its scientific value lies entirely in its experimental falsifiability (Section 8).

4 Derivation of $\tilde{\kappa}$

All results in this section are conditional on Hypothesis 3.1.

4.1 Modified Clausius Relation

Under Hypothesis 3.1, the entanglement entropy contributes an additional term to the Clausius relation. For an entanglement entropy density S_{ent} (bit/m^3), the additional entropy associated with horizon area element dA is:

$$dS_{\text{ent}} = \frac{S_{\text{ent}}}{k_{\text{B}}} \cdot \frac{dV}{4\ell_P}, \qquad dV = \ell_P \, dA, \tag{9}$$

where $\ell_P = \sqrt{\hbar G/c^3} = 1.616 \times 10^{-35}\,\text{m}$ is the Planck length.

Remark 4.1. Equation (9) uses $dV = \ell_P \, dA$, treating one Planck length of depth behind the horizon as the effective thickness of the entanglement contribution. This regulator is borrowed from holographic principles and has no rigorous derivation for non-horizon systems. It is the central unproven assumption of this framework.

4.2 Algebra

Substituting $T_U = \hbar a/(2\pi c k_{\text{B}})$ and $dV = \ell_P \, dA$:

$$\delta Q_{\text{eff}} = \delta Q_{BH} + \frac{\hbar a}{2\pi c k_{\text{B}}} \cdot \frac{S_{\text{ent}}}{k_{\text{B}}} \cdot \frac{dA}{4}. \tag{10}$$

Following Jacobson's identification of $\delta Q_{\text{eff}} = T^{\text{eff}}_{\mu\nu} k^\mu d\Sigma^\nu$, using surface gravity $a = c^2 \kappa_s$, and converting $S_{\text{ent}} \to S_{\text{ent}} \cdot k_{\text{B}} \ln 2$:

$$T^{\text{ent}}_{\mu\nu} = -\frac{c^4}{32\pi G} \, S_{\text{ent}} \, g_{\mu\nu}. \tag{11}$$

Comparing with Eq. (6):

Key Result. Under Hypothesis 3.1 (Planck-scale holographic regulator),

$$\tilde{\kappa} = -\frac{1}{4}. \tag{12}$$

This is an estimate motivated by thermodynamic analogy. It is not a derivation from first principles.

4.3 Environmental Screening

Real systems are continuously decohered by their environments. The effective coupling is:

$$\tilde{\kappa}_{\text{eff}} = -\frac{1}{4} \, \alpha_{\text{screen}}, \tag{13}$$

where $\alpha_{\text{screen}} \in [0, 1]$ is the fraction of the ideal coupling surviving environmental decoherence.

Remark 4.2 (Open Problem 3). The screening factor α_{screen} must be derived from the Lindblad master equation for the specific experimental system. The heuristic range $[10^{-4}, 10^{-2}]$ in earlier drafts applied to few-qubit mesoscopic systems. For macroscopic ensembles ($N \sim 10^6$ atoms, cloud radius $\sim 0.1\,\text{mm}$, coherence length $\sim 1\,\mu\text{m}$), Gaussian spatial screening yields $\alpha_{\text{screen}} \sim 10^{-43}$, making the macroscopic atom-interferometry experiment ill-motivated as the primary near-term target. Until α_{screen} is derived for the relevant regime, the predicted signal magnitude is indeterminate.

5 Radial Scaling: Resolution of OP4

In the Newtonian (weak-field) limit, Eq. (6) becomes a modified Poisson equation for the entanglement potential Ψ:

$$\nabla^2\Psi = \kappa_{\text{eff}}\,\rho_{\text{ent}}(R), \qquad \kappa_{\text{eff}} \equiv \frac{\tilde{\kappa}c^4}{2k_{\text{B}}\ln 2}. \tag{14}$$

Under spherical symmetry, the radial equation (14) integrates to:

$$\Delta a(R) \;\equiv\; -\frac{d\Psi}{dR} \;=\; \frac{\kappa_{\text{eff}}}{R^2}\,M_{\text{ent}}(<R), \tag{15}$$

where the enclosed entropic source is

$$M_{\text{ent}}(<R) = 4\pi\int_0^R \rho_{\text{ent}}(r)\,r^2\,dr. \tag{16}$$

Bounded sources. For a compact atomic ensemble (radius R_0), $M_{\text{ent}}(<R) \to M_{\text{ent}}^{\text{tot}}$ for $R > R_0$, giving:

$$\Delta a(R) \propto \frac{1}{R^2}, \qquad R > R_0. \tag{17}$$

Extended sources. A $1/R$ correction requires $\rho_{\text{ent}}(r) \propto 1/r^2$ so that $M_{\text{ent}}(<R) \propto R$, giving $\Delta a(R) \propto 1/R$. This is *not* a universal consequence of the modified Poisson equation; it encodes a specific density profile assumption.

Remark 5.1. The $1/R$ formula in earlier source documents treated this scaling as universal. Equation (15) resolves the discrepancy: the $1/R$ behavior encodes a hidden assumption that $M_{\text{ent}}(<R) \propto R$. For a compact atomic ensemble, the correct exterior prediction is $1/R^2$. **OP4 is closed.**

6 Current Experimental Bounds

No experiment has been specifically designed to test this coupling. Table 1 summarizes upper bounds from null results in related experiments. The MICROSCOPE bound is omitted pending derivation of the explicit mapping from satellite observables to $\tilde{\kappa}$ (see Section 6.1).

Table 1: Experimental upper bounds on $|\tilde{\kappa}|$ from null results.

Experiment	Constraint	Reference
Gravity-mediated entanglement	$|\tilde{\kappa}| < 3 \times 10^{-9}$	Bose et al. [3]†
Atom interferometry (dual-species)	Constrains η at 10^{-12}	Asenbaum et al. [4]

†Mapping from gravity-entanglement bound to $\tilde{\kappa}$ bound to be verified.

6.1 MICROSCOPE and the Missing Mapping

MICROSCOPE constrains the Eötvös parameter at the 10^{-15} level [5]. Both test masses are classical solids with $S_{\text{ent}} \approx 0$: the entanglement source term contributes essentially equally (zero) to both trajectories, yielding $\eta \approx 0$ independently of $\tilde{\kappa}$. *A MICROSCOPE-derived bound on $\tilde{\kappa}$ requires an explicit argument mapping microscopic entanglement structure to satellite-observable gravitational anomalies. Until that argument is constructed, no MICROSCOPE bound on $\tilde{\kappa}$ is stated.*

7 Revised Experimental Protocol: NISQ-Scale Test

7.1 Motivation for the Revised Target

The original Section 5 proposed $N \geq 10^6$ ^{87}Rb atoms in a GHZ-like state. Given $\alpha_{\text{screen}} \sim 10^{-43}$ for a macroscopic BEC (Section 4), this target is not well-motivated as the primary near-term experiment. We revise the proposal to a 50–100 qubit NISQ-scale platform.

7.2 Platform Specifications

Table 2: Minimum NISQ platform requirements for the revised experiment.

Parameter	Requirement	Current capability (2025)
Qubit count	50–100	$>$ 100 (multiple platforms)
Entanglement depth	$\geq$ 10 qubits (verified)	Achievable with 2-qubit gates
Gate fidelity (2Q)	$\geq$ 99%	99–99.5% (best platforms)
Coherence time	$\geq 100\,\mu$s	$100\,\mu$s–s
State verification	Witnesses or tomography	Standard

7.3 Protocol

Branch A (entangled). Prepare an N-qubit state with verified entanglement depth $k \geq 10$: GHZ-like, cluster state, or spin-squeezed state. Verify entanglement using witnesses and subsystem purity.

Branch B (reference). Apply controlled dephasing (random single-qubit Z rotations) to destroy inter-qubit entanglement while preserving qubit count, spatial geometry, and gross energy budget. Verify entanglement depth ≤ 1 (product state).

Observable. Measure a differential observable (phase shift, frequency shift, or effective acceleration proxy) that could be sensitive to an entanglement-dependent coupling. Systematically vary entanglement depth between runs and look for any reproducible dependence.

7.4 Signal Magnitude

The predicted signal magnitude is indeterminate. The structure of the signal is given by Eq. (15); any numeric forecast requires first resolving α_{screen} (Open Problem 3) and the Planck-scale renormalization of $\tilde{\kappa}$ (Open Problem 10). The experiment is designed not to claim detectability but to produce a bound on $|\tilde{\kappa}_{\text{eff}}| \cdot M_{\text{ent}}^{\text{tot}}/R^2$ — a meaningful experimental result regardless of outcome.

7.5 Systematic Error Budget

Table 3: Dominant systematics in the NISQ-scale protocol and their mitigations.

Systematic	Origin	Mitigation
Imperfect state prep.	Gate errors, init. fidelity	Entanglement verification per run
Decoherence drift	Time-dependent noise	Alternating A-B-A-B; calibration
Geometry coupling	Device layout	Model $\rho_{\text{ent}}(x)$ per architecture
Platform cross-talk	Qubit–qubit coupling	Symmetrized pulses; null circuits
Ambiguous screening	α_{screen} unknown	State signal as indeterminate

7.6 External Benchmarks

The 2020 Asenbaum et al. experiment [4] used a dual-species ^{85}Rb/^{87}Rb interferometer with $T \sim 2\,\text{s}$ free fall:

$$\eta = [1.6 \pm 1.8\,(\text{stat}) \pm 3.4\,(\text{syst})] \times 10^{-12}, \tag{18}$$

with per-shot resolution $\sim 1.4 \times 10^{-11}\, g$. Any entanglement-induced effect behaving as a composition-dependent fifth force must respect this bound.

8 The Falsification Criterion

The falsification criterion is independent of the signal magnitude prediction and holds regardless of how α_{screen} and the Planck-scale renormalization are resolved.

Key Result. Falsification conditions F1–F5. The framework is falsified for laboratory-scale relevance if *all* of the following hold simultaneously:

F1 Quantum-coherent systems with $\geq 10^6$ genuinely entangled qubits are achieved, with entanglement depth $\geq 10^3$ verified by witnesses.

F2 Pressure sensitivity reaches $\Delta p < 10^{-6}$ Pa.

F3 At least 1000 independent experimental runs yield statistically consistent null results.

F4 Null results are confirmed on ≥ 3 independent platforms (e.g., trapped neutral atoms, superconducting qubits, optomechanical oscillators).

F5 No stress-energy contribution beyond standard decoherence models is detected at the F2 sensitivity level.

Consequence: $|\tilde{\kappa}| < 10^{-15}$, ruling out laboratory-scale relevance for the foreseeable future.

Why this criterion is strong. It is quantitative, platform-independent, statistically rigorous, achievable with current and near-term technology, and admits no escape hatches (entanglement depth and sensitivity are fully specified). A null result satisfying F1–F5 is not a failure — it is a precise bound on a previously unmeasured quantity.

9 Open Problems

OP3 (Critical): Lindblad derivation of α_{screen}. Derive α_{screen} from a microscopic open-quantum-system model with explicit Hamiltonian, bath, and coupling. Without this, predicted signal sizes are indeterminate and cannot be compared against any experimental bound. The macroscopic BEC calculation ($\alpha_{\text{screen}} \sim 10^{-43}$) and the NISQ heuristic ($\alpha_{\text{screen}} \sim 10^{-4}$–$10^{-2}$) may reflect regime differences rather than contradiction; a Lindblad derivation would resolve which regime applies.

OP8 (High): Bianchi identity compatibility. The contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ implies $\nabla^\mu T^{\text{matter}}_{\mu\nu} = -(\tilde{\kappa} c^4/8\pi G k_{\text{B}} \ln 2)\nabla_\nu S_{\text{ent}}$. Standard matter is therefore not independently energy-momentum conserved when $\nabla S_{\text{ent}} \neq 0$. Estimate the magnitude of this

anomalous force for the proposed NISQ experiment and compare against precision tests of energy-momentum conservation.

OP10 (Critical, long-term): Planck-scale renormalization. The coupling prefactor $c^4/(8\pi k_{\mathrm{B}} \ln 2) \approx 5 \times 10^{65}$ in SI units reflects the Planck-scale origin of the derivation. Whether $\tilde{\kappa} = -1/4$ at the Planck scale renormalizes to a far smaller effective value at laboratory scales — and what the renormalized value is — is an open problem in quantum gravity phenomenology.

10 Discussion and Conclusion

10.1 Relationship to Established Physics

This framework is conservative in ontological commitments: classical GR unchanged, standard quantum mechanics unchanged, no new particles or fields. The only novel claim is that macroscopic quantum entanglement sources stress-energy in a way not captured by the standard $T_{\mu\nu}$.

10.2 Relationship to Other Programs

vs. ER=EPR: Maldacena and Susskind [6] propose entanglement is equivalent to geometric connectivity (wormholes). The present framework proposes entanglement *sources* stress-energy. These are complementary, not competing.

vs. Verlinde emergent gravity: Verlinde [2] proposes gravity as an entropic force at galactic scales. The present framework proposes a laboratory-scale realization with a specific experimental test.

10.3 Conclusion

We have constructed a dimensionally consistent framework in which quantum entanglement entropy density sources spacetime curvature. The coupling constant $\tilde{\kappa} = -1/4$ is estimated under a Planck-scale holographic regulator hypothesis that is motivated but unproven. The signal magnitude is indeterminate pending resolution of the screening factor. The falsification criterion is precise, platform-independent, and achievable with near-term technology.

The framework makes no claim that a signal is detectable. It claims only: either an entanglement-dependent gravitational effect will be observed under conditions F1–F5, or the framework is ruled out for laboratory-scale relevance. Both outcomes constitute scientific progress.

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